Practice Sheet # 6

Successive Differentiation

a. Find the *n* th derivative of the following functions:

1. $y = x^n$
2. $y = (ax+b)^n$
3. $y = \ln(ax+b)$
4. $y = \dfrac{1}{x+a}$
5. $y = e^{ax}$
6. $y = \sin(ax+b)$
7. $y = \cos(ax+b)$

b. If $y = e^{ax}\sin bx,$ then show that $y_2 - 2ay_1 + (a^2+b^2)y = 0.$

c. If $y = e^x \sin x,$ then show that $y_4 + 4y = 0.$

Answers

a) 1. $n!$

2. $n!\,a^n$

3. $(-1)^{n+1}\frac{(n-1)!a^n}{(ax+b)^n}$

4. $\frac{(-1)^n n!}{(x+a)^{n+1}}$

5. $a^n e^{ax}$

6. $(-1)^{\left\lfloor\frac{n}{2}\right\rfloor}a^n\begin{cases}\sin(ax+b) & \text{if } n \text{ is even}\\ \cos(ax+b) & \text{if } n \text{ is odd}\end{cases}$

OR $a^n\sin\left(n\cdot\frac{\pi}{2}+ax+b\right)$

7. $(-1)^{\left\lceil\frac{n}{2}\right\rceil}a^n\begin{cases}\sin(ax+b) & \text{if } n \text{ is odd}\\ \cos(ax+b) & \text{if } n \text{ is even}\end{cases}$

OR $a^n\cos\left(n\cdot\frac{\pi}{2}+ax+b\right)$